

FIT3003 Major Assignment 2 - S2 2025 (Weight = 40%)

Due date: Friday, 17 October 2025, 11:55 PM

Version: 1.2 – 18/09/2025

General Information and Submission

- This is an individual assignment.
- This assignment requires the Oracle database.
- *Submission method*: Submission is online through Moodle.
- *Penalty for late submission*: 5% deduction for each day.
- *Assignment FAQ*: There is an [Assignment Frequently Asked Questions page](#) set up for the Assignment 2 on EdStem Forum.
- A marking rubric has been provided for Assignment 2: [A2 Marking guide](#)

Problem Description

Monash Equipment Centre (MonEquip) is an Australian-based company that provides a wide range of equipment for civil infrastructure and commercial construction projects. The company has two main business functions:

- selling equipment along with excellent after-sales service,
- providing equipment for hire while ensuring a seamless hiring experience.

MonEquip has different branches throughout the state of Victoria, in which the management team frequently generates reports to keep track of the business (e.g. calculating the revenues from hiring and selling equipment). The reports are then used for forecasting various trends and making predictions about the business.

MonEquip currently has an existing operational database which maintains and stores all of the information of their business transactions (e.g. product hiring, product sales, equipment, etc.) required for the management's daily operation. The operational database records the transaction **from April 2018 to December 2020**. However, since the staff at MonEquip has limited database knowledge and the operational database is quite large, the management team has decided to hire you as a Data Warehouse Engineer to design, develop, and quickly generate Business Intelligence reports from a new Data Warehouse.

As you are required to design a Data Warehouse for **MonEquip**, the Operational Database tables can be accessed through the [MonEquip](#) account. You can, for example, execute the following query:

Select * From MonEquip.<table_name>;

The data definition of each table in MonEquip is as follows:

Table Name	Attributes and Data Types		Notes
ADDRESS	ADDRESS_ID	NUMBER	This table contains customer address information
	STREET_NUMBER	NUMBER	
	STREET_NAME	VARCHAR2	
	SUBURB	VARCHAR2	
	STATE	VARCHAR2	
	POSTCODE	NUMBER	
CATEGORY	CATEGORY_ID	NUMBER	This table contains category information
	CATEGORY_DESCRIPTION	VARCHAR2	
CUSTOMER	CUSTOMER_ID	NUMBER	This table contains customer information
	CUSTOMER_TYPE_ID	NUMBER	
	NAME	VARCHAR2	
	GENDER	VARCHAR2	
	ADDRESS_ID	NUMBER	
	PHONE	VARCHAR2	
	EMAIL	VARCHAR2	
CUSTOMER_TYPE	CUSTOMER_TYPE_ID	NUMBER	This table contains customer type information
	DESCRIPTION	VARCHAR2	
EQUIPMENT	EQUIPMENT_ID	NUMBER	This table contains equipment information
	EQUIPMENT_NAME	VARCHAR2	
	EQUIPMENT_PRICE	NUMBER	
	MANUFACTURE_YEAR	NUMBER	
	MANUFACTURER	VARCHAR2	

	CATEGORY_ID	NUMBER	
HIRE	HIRE_ID	NUMBER	This table contains hire information
	START_DATE	DATE	
	END_DATE	DATE	
	EQUIPMENT_ID	NUMBER	
	QUANTITY	NUMBER	
	UNIT_HIRE_PRICE	NUMBER	
	TOTAL_HIRE_PRICE	NUMBER	
	CUSTOMER_ID	NUMBER	
	STAFF_ID	NUMBER	
SALES	SALES_ID	NUMBER	This table contains sales information
	SALES_DATE	DATE	
	EQUIPMENT_ID	NUMBER	
	QUANTITY	NUMBER	
	UNIT_SALES_PRICE	NUMBER	
	TOTAL_SALES_PRICE	NUMBER	
	CUSTOMER_ID	NUMBER	
	STAFF_ID	NUMBER	
STAFF	STAFF_ID	NUMBER	This table contains the information of staff
	FIRST_NAME	VARCHAR2	
	LAST_NAME	VARCHAR2	
	GENDER	VARCHAR2	
	PHONE	VARCHAR2	
	EMAIL	VARCHAR2	
	COMPANY_BRANCH	VARCHAR2	

For the daily operation, MonEquip normally purchases equipment from its suppliers. The company then resells the equipment to customers while also providing a hiring service.

When an equipment is purchased, the price of the equipment is kept in the Equipment table as EquipmentPrice.

Meanwhile, the hiring rate is calculated as follows:

- The unit hire price is the hiring rate per day.
- If the customer returns the equipment within the same day, they only need to pay for 50% of the *Unit_Hire_Price*.
- Otherwise, *Total_hire_price* is calculated as $(End_Date - Start_Date) * Unit_Hire_Price * Quantity$.

A. Transformation Stage

The first stage of this assignment is divided into **TWO** main tasks:

1. Design a data warehouse for the above MonEquip database.

You are required to create a data warehouse for the **MonEquip** database. The management is especially interested in the following indicators :

- Average hire revenue per hire transaction
- Average sales revenue per sale transaction
- Total revenue for equipment hiring
- Total revenue for sales
- Number of equipment hired
- Number of equipment sold

Note:

In the MonEquip database, the *Total Sales Price* and *Total Hire Price* fields represent the total amount received from each sales or hire transaction. For this assignment, you can use these fields to calculate revenue values when designing your data warehouse.

While the database uses the term *price*, it's common in real-world systems for attribute names in operational databases to differ from the terminology used in business reporting or analytics. *Operational systems may use technical or transactional labels, whereas analytical systems often adopt terms that reflect business meaning more clearly—such as referring to **total prices** as **forms of revenue**.*

As part of the warehouse design process, you are expected to make informed decisions about how to interpret and organize such fields to support business-focused analysis.

The following shows some possible dimension attributes that you may need in your data warehouse:

- Time: Month, Year (derived from the **Start Date of Hire** for hire transactions and the **Sale Date** for sales transactions)
- Season [Spring: 9 to 11, Summer: 12 to 2, Autumn: 3 to 5 and Winter: 6 to 8]
- Customer type
- Category
- Company branch
- Sales price scale: low sales <\$5,000; medium sales between \$5,000 and \$10,000; high sales > \$10,000

For the attribute, ensure that it meets the requirements of the range or group specified in your submission, if required in the specification.

- **Preparation stage.**

Before you start designing the data warehouse, you have to ensure that you have explored the operational database and have done sufficient data cleaning. Once you have done the data cleaning process, you are required to explain what strategies you have taken to explore and clean the data.

The outputs of this task for **Report** are:

- a) If you have done the data cleaning process, explain the strategies you used in this process (you need to show the SQL to explore the operational database and SQL of the data cleaning, as well as the screenshot of data *before* and *after* data cleaning).

- **Designing the data warehouse by drawing star/snowflake schema.**

Design task A:

The star schema for this data warehouse may contain multi-facts. You need to identify the fact measures, dimensions, and attributes of the star/snowflake schema. The following queries might help you to determine the fact measures and dimensions:

- What was the total sales revenue in January, 2020?
- How many pieces of equipment were sold in Winter, 2018?
- How many equipment was hired by business customers?
- What was the total hire revenue in Clayton branch?
- How many trailers were hired by individual customers in Summer?
- What is the average sale revenue for Lighting equipment in 2019?
- What is the average hire revenue for Vehicles by individual customers?
- How much sales revenue was generated from a high sale in Summer?

Note:

- The star schema you created in Design Task A as the highest level of aggregation.
- The star or snowflake schema you design may or may not require a **bridge table**. If a bridge table is necessary, make sure to include the attributes **GroupList** and **WeightFactor** in the relevant dimension table.

Design task B:

In this assignment, consider the star schema you created in Design Task A as the highest level of aggregation. The MonEquip company manager wants to implement a drill-down function to explore more detailed information. Your task is to **suggest several ways** to increase the granularity of your fact tables from Design Task A. In other words, the manager wants to decrease the aggregation level of the fact tables you created in Design Task A.

The outputs of task A & B for **Report** are:

- b) A star/snowflake schema diagrams for design task A. (You can use [Lucidchart](#) to draw the star schema.)
- c) List suggestion of increase the granularity of your fact tables for design task B
- d) An explanation of the difference among SCD types 1, 2, 3, 4, and 6. Provide an example of a temporal dimension and describe which SCD type(s) would be appropriate for that dimension in Design Task A or Design Task B. If your design includes a temporal dimension, you may explain that instead.

Note:

The star/snowflake schema design, including the fact measures and dimensions, is entirely based on the specific questions we provide for each exercise and assignment. It is essential to closely follow the question requirements and design the star schema accordingly. The fact measures and dimensions should align precisely with the data and analysis needed to address the given problem.

Please ensure that you thoroughly understand the question prompts and the expected outcomes before proceeding with the star schema design. **Any deviation from the provided question may lead to inaccurate results and impact your assessment.**

Asking the tutor to check whether your design is correct or not will not be answered, as it is part of the assessment in this assignment.

2. Implement **design task A star/snowflake schema using SQL.**

You are required to implement the star/snowflake schema that you have drawn in **design task A**. This implies that you need to create the fact and dimension tables in SQL. The

output is a series of SQL statements to perform this task. You will also need to show that this task has been carried out successfully.

Note:

- If your account is full, you will need to drop all of the tables that you have previously created during the tutorials.
- If you have dropped all tables in your account and you still encounter the ORA-01536: space quota exceeded for tablespace 'TABLE_NAME', please check your SQL code whether you have properly joined all tables. This issue was mainly caused when you did not do the table join properly as the number of records multiplied during the process.

The outputs of this task for **Report** are:

- a) Screenshots of the table structure you created for Design Task A, including the dimension tables and fact tables.

A sample of screenshots of the table structure

Address Table

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ADDRESS_ID	NUMBER	Yes	(null)	1	(null)
2	STREET_NUMBER	NUMBER	Yes	(null)	2	(null)
3	STREET_NAME	VARCHAR2(100 BYTE)	Yes	(null)	3	(null)
4	SUBURB	VARCHAR2(20 BYTE)	Yes	(null)	4	(null)
5	STATE	VARCHAR2(10 BYTE)	Yes	(null)	5	(null)
6	POSTCODE	NUMBER	Yes	(null)	6	(null)

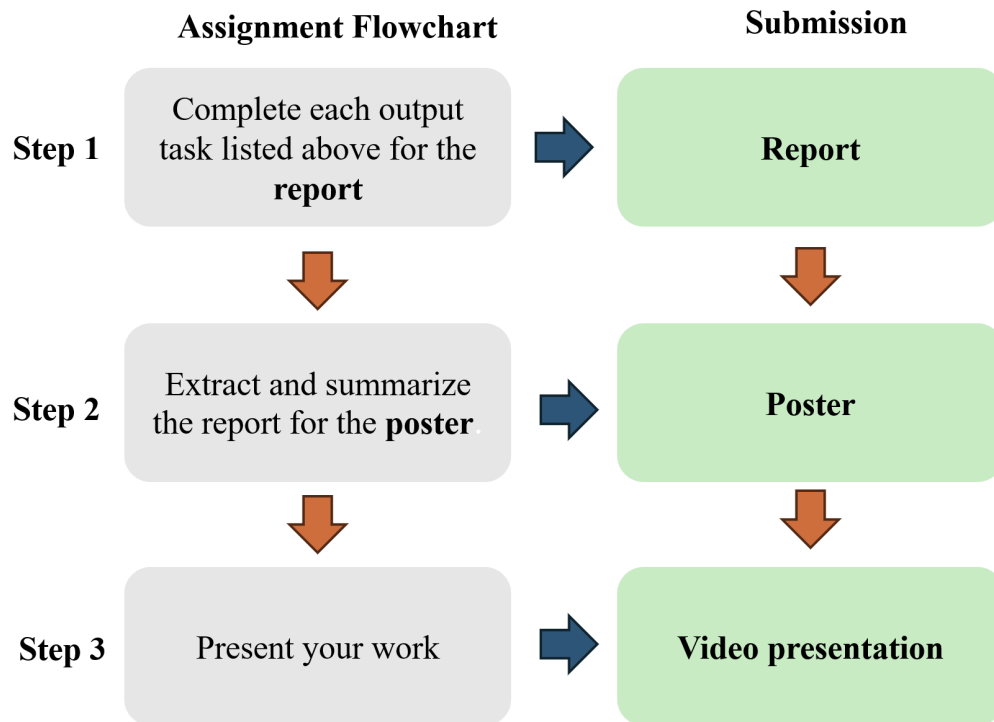
B. Data Analytic Stage

Conduct a data analysis using the star schema you created in Design Task A by writing SQL queries to explore the data further. Present your findings in a clear and concise manner, demonstrating your understanding of the dataset and highlighting any noteworthy observations or patterns.

The outputs of this task for **Report** are:

1. Findings report: A detailed explanation of your findings, including any significant observations or patterns identified during the analysis.

Submission Checklist



Step 1: Report (30% of the total score)

A combined pdf file save as: YourstudentID_A2_report.pdf, containing all of the above tasks:

- A. Cover page
- B. If you have done the data cleaning process, explain the strategies you used in this process (you need to show the SQL to explore the operational database and SQL of the data cleaning, as well as the screenshot of data *before* and *after* data cleaning). Note that you are only required to find around 5 (five) data errors for this stage.
- C. A star/snowflake schema diagrams for design task A
- D. List suggestion of increase the granularity of your fact tables for design task B
- E. An explanation of the difference among SCD types 1, 2, 3, 4, and 6. Provide an example of a temporal dimension and describe which SCD type(s) would be appropriate for that dimension in Design Task A or Design Task B.
- F. Screenshots of the table structure you created for Design Task A only, including the dimension table and fact tables.
 - a. SQL file for creating the star/snowflake schema is **NOT** required in submission
- G. Findings report: A detailed explanation of your findings, including any significant observations or patterns identified during the analysis.

Step 2: Poster (35% of the total score)

One page standard A4 poster in PDF format to save as: YourstudentID_A2_poster.pdf

Extract key information from the report you created and present it in a **one-page poster**. The poster must be in **standard A4 size and in PDF format**, which can be either landscape or portrait. The content should be clear and easy to understand. Avoid using technical jargon or complex language. Review the poster before submission to ensure it effectively communicates the key messages of your report.

Note:

Ensure the poster content is consistent with the key structure and findings of your report, and choose an appropriate layout that effectively organizes the information in a clear and logical manner. Maintain a good balance of text and visuals to enhance readability, and ensure all visuals are relevant and support the content of the poster. Label all visuals clearly and provide captions where necessary. Avoid overcrowding the poster with too much text or too many visuals, and ensure the poster is free of any grammatical or typographical errors.

Key guidance of design a poster:

- What is the main theme/objective of the poster that you want to express?
- Who is your target audience for this poster?
- Do you really need all the details from your report on this poster?

Step 3: Video presentation (35% of the total score)

A five minute video presentation in mp4 format save as: YourstudentID_A2_video.mp4

Based on the report and poster you have created, present your design and findings in a **five-minute video presentation**. Ensure you thoroughly understand both the report and the poster to effectively extract and communicate the key points.

Assignment Submission

The assignment must be submitted electronically through Moodle. Please ensure the following:

1. Step 1 output: A combined pdf file save as: YourstudentID_A2_report.pdf
2. Step 2 output: One page standard A4 poster in PDF format to save as: YourstudentID_A2_poster.pdf

3. Step 3 output: A five minute video presentation in mp4 format save as: YourstudentID_A2_video.mp4

Zip all above files from step 1 to 3, and name the ZIP folder as A2_YourstudentID.zip.

The due date: **Due - Friday, 17th October 2025, 11:55 PM**

- The submission of this assignment must be in the form of a **single ZIP file**. Only PDF and .mp4 files will be accepted within the zip file. No other formats will be accepted.
- You must ensure that you have all the files listed in this checklist before submitting your assignment to Moodle. Failure to submit a complete list of files will lead to mark penalties.
- It is important to note that our support hours are limited and we don't have the capacity to deal with submission issues outside of working hours. You must ensure that you have all the files listed in this checklist before submitting your assignment to Moodle. Failure to submit a complete list of files will result in a mark penalty.
- Penalty for late submission: 5 % deduction for each day, including weekends.
- Submission Cut-off time: **24th October 2025, 11:55 PM** (Submission link will be unavailable after the cut-off date).

Authorship

This assignment is an **individual assignment** and the final submission must be identifiably your own work. Breaches of this requirement will result in an assignment not being accepted for assessment and may result in disciplinary action.

Late Penalty:

Late assignments submitted without an approved extension may be accepted up to a maximum of **seven days** with the approval of the Chief Examiner and/or Lecturer but will be **penalised at the rate of 5% per day (including weekends and public holidays)**. Assignments submitted more than seven days after the due date will receive a zero mark for that assignment and may **not receive any feedback**.

Please note:

- An inability to manage your time or computing resources will not be accepted as a valid excuse. (Several assignments being due at the same time are a fact of university life.)
- Hardware failures, whether of personal or university equipment, are not normally recognised as valid excuses. Failure to back up assignment files is also not recognised as a valid excuse.

Special Consideration:

All extensions / special considerations will now be handled by the central Spec Con team. **Please do not email teaching staff to request an extension or special consideration.**

Extensions and other individual alterations to the assessment regime will only be considered using the University Special Consideration Policy. Students should carefully read the [Special Consideration website](#), especially the details about what formal documentation is required.

All special consideration requests should be made using the [Special Consideration Application](#).

Please do not assume that submission of a Special Consideration application guarantees that it will be granted – you must receive an official confirmation that it has been granted.

Getting help and support:

What can you get help for?

- ***Consultations with the Teaching Team***

Talk to the Teaching Team:

<https://learning.monash.edu/course/view.php?id=19675§ion=5>

- ***English language skills***

Talk to English Connect: <https://www.monash.edu/english-connect>

- ***Study skills***

Talk to a learning skills advisor: <https://www.monash.edu/library/skills/contacts>

- ***Counselling***

Talk to a counsellor: <https://www.monash.edu/health/counselling/appointments>

Plagiarism and Collusion:

Monash University is committed to upholding standards and academic integrity and honesty. Please take the time to view these links.

[Academic Integrity Module](#)

[Student Academic Integrity Policy](#)

[Test your knowledge, collusion \(FIT No Collusion Module\)](#)

All the best for your Assignment!